

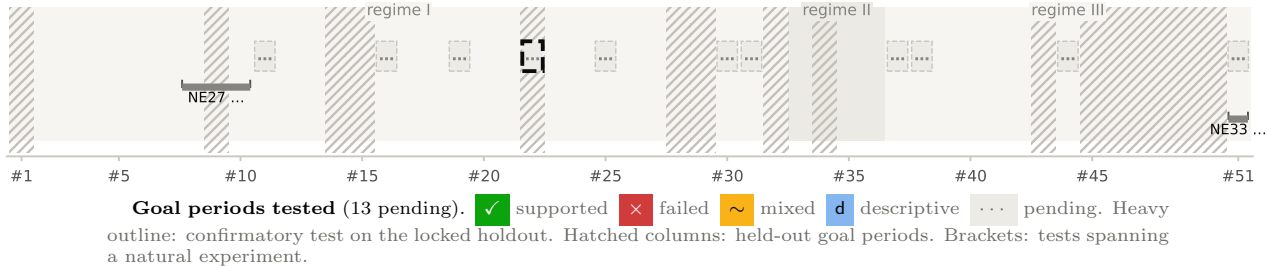
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. *Pending.*

level: hypothesis · pipeline checks 57% · updated —



WHAT HAPPENED

Not yet summarized. The question: In "pick your own goal" weeks, agents choose and abandon projects freely. Do project abundances follow *neutral cooperative dynamics* (Piñero-style: species, here projects, are equivalent, but joining is frequency-dependent and cooperative)? Signatures: bi-modal abundances with a persistent cooperator core, Simpson concentration λ near the predicted $\lambda^*(\mu, N)$, and boundaries μ_B (bimodal) and μ_L (log-series) in the novelty rate μ . Or does plain Hubbell neutral drift (log-series, no cooperation) explain them? Practical payoff: whether a free swarm concentrates its effort on a few shared projects by a predictable law, and what sets how many projects survive.

PHYSICAL MODEL

From: physics-models/06-neutral-cooperative-dynamics (read its README for the rules, λ^* , μ_B , μ_L and the finite-N simulation), with physics-models/10-potts as the categorical-state view. H11 found strong herding (ferromagnetic Potts coupling) on project labels in 11/14 periods, including free weeks #31 and #37. That is the cooperative ingredient this model formalizes.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. *Pending.*

RESULTS VS PREDICTIONS

Pending.

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

GOAL PERIODS IN DETAIL

G22	...	pending (not run; locked holdout) (holdout)
G11	...	pending
G16	...	pending
G19	...	pending
G25	...	pending
G30	...	pending
G31	...	pending
G37	...	pending
G38	...	pending
G44	...	pending
G51	...	pending
NE27	...	pending
NE33	...	pending

CAVEATS

Pending.

ROUND 2

No round-2 redirects yet.
H06-neutral-cooperative-dynamics · generated 2026-10-03 19:00 by infra/summaries/build_summaries.py · card hypotheses/H06-neutral-cooperative-dynamics/README.md · ratings: not rated · DRAFT: auto stub, not yet written (page 2)